```
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// Searching contact name using Sequential and Binary Search


#include<stdio.h>
#include<string.h>


typedef struct contact{
    char name[20];
    char phone[20];
    char address[20];
}contact;


void main(){
    contact c[20];
    int n, i;
    int lincom = 0;
    int linfound = 0;
    int bincom = 0;
    char key[20];

    printf("Enter number of contacts: ");
    scanf("%d", &n);


    printf("\n---//ENTER CONTACT DETAILS (A-Z)//---\n");
    for(i = 0; i < n; i++){
```

```
        printf("\n~~~Contact %d~~~\n", i+1);

        printf("Name: ");

        scanf("%s", &c[i].name);

        printf("Contact: ");

        scanf("%s", &c[i].phone);

        printf("Address: ");

        scanf("%s", &c[i].address);

    }


    printf("\n---//DISPLAYING CONTACTS//---\n");

    for(i = 0; i < n; i++){

        printf("\n~~~Contact %d~~~", i+1);

        printf("\nName: %s", c[i].name);

        printf("\nContact: %s", c[i].phone);

        printf("\nAddress: %s", c[i].address);

    }

    printf("\n\n---//END//---\n\n");


    printf("Enter name of contact to search: ");

    scanf("%s", &key);

    printf("\n---//Linear Search//---");

    for(i = 0; i < n; i++){

        lincom++;

        if(strcasecmp(key, c[i].name)==0){

            printf("\nCONTACT FOUND");
```

```
printf("\nName: %s", c[i].name);

printf("\nContact: %s ", c[i].phone);

printf("\nAddress: %s", c[i].address);

linfound = 1;

break;

        }

}

if(linfound == 0){

        printf("\nCONTACT NOT FOUND");

}



printf("\nLinear comparisons: %d", lincom);

printf("\n\n---//Binary Search//---");

//Sorting Contact List to apply binary search

char temp[20];

strcpy(temp, c[0].name);

for(int i = 0; i < n; i++){

        for(int j = 0; j <    n-i-1; j++){

                if(strcasecmp(c[j].name, c[j+1].name) > 0){

                        strcpy(temp, c[j].name);

                        strcpy(c[j].name, c[j+1].name);

                        strcpy(c[j+1].name, temp);

                }

        }

}
```

```
int high = n-1;

int low = 0;

int mid;

while(low <= high){

    mid = (high+low)/2;

    if(strcasecmp(c[mid].name, key) == 0){

        bincom++;

        printf("\nContact found at location %d", mid+1);

        break;

    }else if(strcasecmp(c[mid].name, key)<0){

        bincom++;

        low = mid+1;

    }else{

        bincom++;

        high = mid-1;

    }

}

if(low>high){

    printf("\nContact not found");

    }


    printf("\nNumber of binary comparisons: %d", bincom);


}
```

//Output

Enter number of contacts: 5

---//ENTER CONTACT DETAILS (A-Z)//---

~~~Contact 1~~~

Name: Neel

Contact: 9874125630

Address: Satara

~~~Contact 2~~~

Name: Abhishek

Contact: 9632014587

Address: Nagpur

~~~Contact 3~~~

Name: Pushpak

Contact: 9874563210

Address: Pune

~~~Contact 4~~~

Name: Utkarsh

Contact: 8520147963

Address: Mumbai

~~~Contact 5~~~

Name: Chaitanya

Contact: 5203698741

Address: Pune

---//DISPLAYING CONTACTS//---

~~~Contact 1~~~

Name: Neel

Contact: 9874125630

Address: Satara

~~~Contact 2~~~

Name: Abhishek

Contact: 9632014587

Address: Nagpur

~~~Contact 3~~~

Name: Pushpak

Contact: 9874563210

Address: Pune

~~~Contact 4~~~

Name: Utkarsh

Contact: 8520147963

Address: Mumbai

~~~Contact 5~~~

Name: Chaitanya

Contact: 5203698741

Address: Pune

---//END//---

Enter name of contact to search: utkarsh

---//Linear Search//---

CONTACT FOUND

Name: Utkarsh

Contact: 8520147963

Address: Mumbai

Linear comparisons: 4

---//Binary Search//---

Contact found at location 5

Number of binary comparisons: 3